SPECIAL ISSUES ON VALUATION (3/3)

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20127 Course on Corporate Finance (Business Valuation), MSc in Finance, Bocconi University

Application of Put-Call Parity to the Valuation of Risky Debt

Module structure

1 **Put-call parity**

2 **Application of put-call parity to the valuation of risky debt**

Put-call parity

Portfolio

- Buy (+) the equity at spot price S
- Sell (-) a call on S with strike X and maturity T
- Buy (+) a put on S with the same strike X and maturity T

$$+S - C(S,X,T) + P(S,X,T)$$

$$= B(X,T)$$

The portfolio pays X at maturity in every state. B is a risk-free zero-coupon bond with face value X.

$$+S = +C + B - P$$

How put-call parity works

Example: strike X = 40 and two scenario of St at maturity (higher and lower than X)

Position	$S_t = 45$	$S_t = 35$
Equity S	45	35
Short call −C	−5	0
Long put +P	0	+5
Portfolio value	40	40

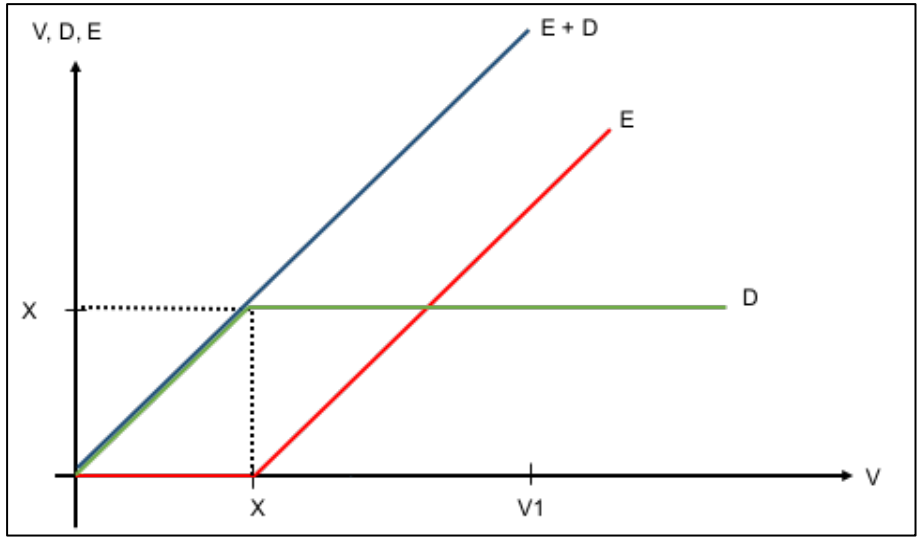
Both scenarios deliver the strike price X = 40.

From put-call parity to risky debt

V	Levered value of the corporation’s assets
C_v	Call on V with strike equal to debt face value X and maturity T
P_v	Put on V with strike X and maturity T
B	Risk-free zero-coupon bond with face value X and maturity T

The risk profiles of equity and debt

State	Equity E	Risky debt D
$V_t \leq X$	0	V_t
$V_t > X$	$V_t - X$	X

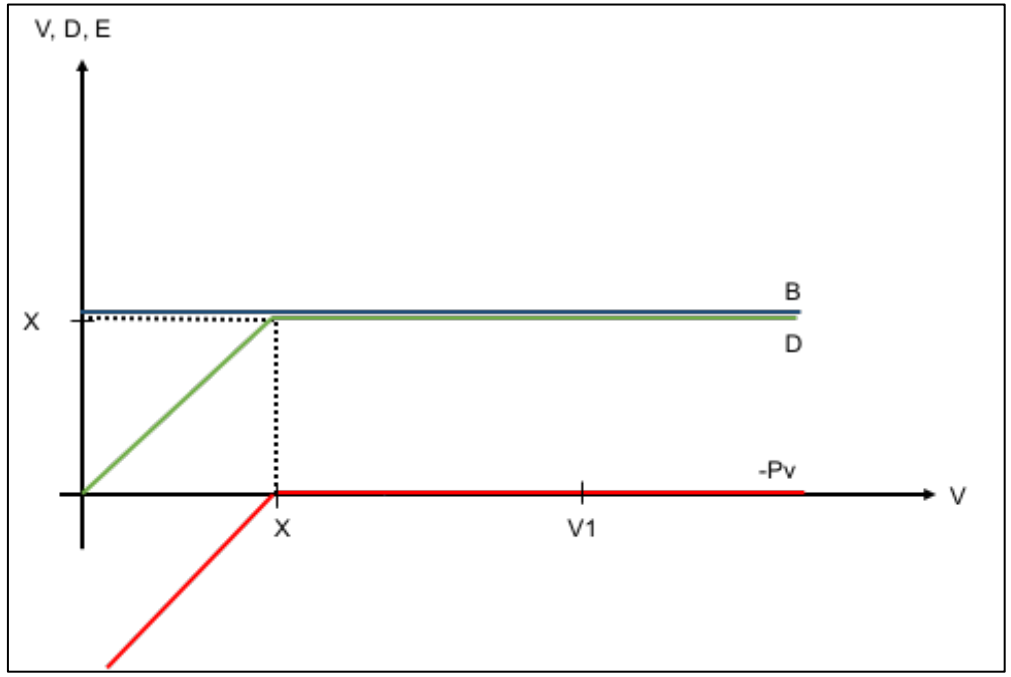


$$V = C_V + \left(B - P_V\right)$$

E D

Below X, debt absorbs changes in firm value. Above X, equity captures the upside.

The risk profile of risky debt



$$D = B - P_v$$

- B is risk-free debt.
- $-P_v$ is the put sold to equity holders.
- $B - P_v$ is the risk profile of the risky debt

From risk profile to risky-debt pricing

- The bond holder:
  - buys a risk-free zero-coupon bond B with face value X and maturity T;
  - sells a put P_v to equity holders, who can deliver the firm when $V_t \leq X$.

- Higher leverage or asset volatility raises P_v, lowers D and raises the promised yield.

$$D = B - P_v$$

The option model prices contractual debt cash flows and therefore produces a promised return.

Promised return and expected return (a simplified example with some assumpions)

Debt market value today	**88**
Promised payment in one year	**100**
Default probability	**10%**
Recovery in default	**40**
Expected payoff	**94**
Promised return	**13.64%**
Expected return	**6.82%**

Expected return uses the probability-weighted payoff, not the contractual face value.

The WACC requires an expected cost of debt

k_D, expected ≈ promised yield – expected-loss yield

- The option-pricing model returns a promised yield because it uses contractual repayment.
- The CAPM cost of equity is an expected return that compensates only for non-diversifiable risk.
- Using promised yield in WACC can overstate k_D, overstate WACC and understate enterprise value.

Use a consistent expected-return convention for both debt and equity.

Review questions

IN PRACTICE Answer through the economic identity, not by memorising the option sign.

Q1 To obtain risky-debt value under put-call parity, the put premium should be:

- Subtracted from the value of risk-free debt
- Added to the value of risk-free debt
- Multiplied by the value of risk-free debt

Q2 The cost of debt included in WACC should equal:

- The promised yield
- The expected return: promised yield minus the yield due to expected default loss
- The expected return: promised yield plus the yield due to expected default loss

Solutions

IN PRACTICE Explain why each answer is correct.

Q1

Subtract the put premium from risk-free debt

$D = B - P_v$

Q2

Use expected return in WACC

Promised yield minus expected-default-loss yield

Using promised yield can overstate WACC and understate corporate value.